# Digital Twin Final Project

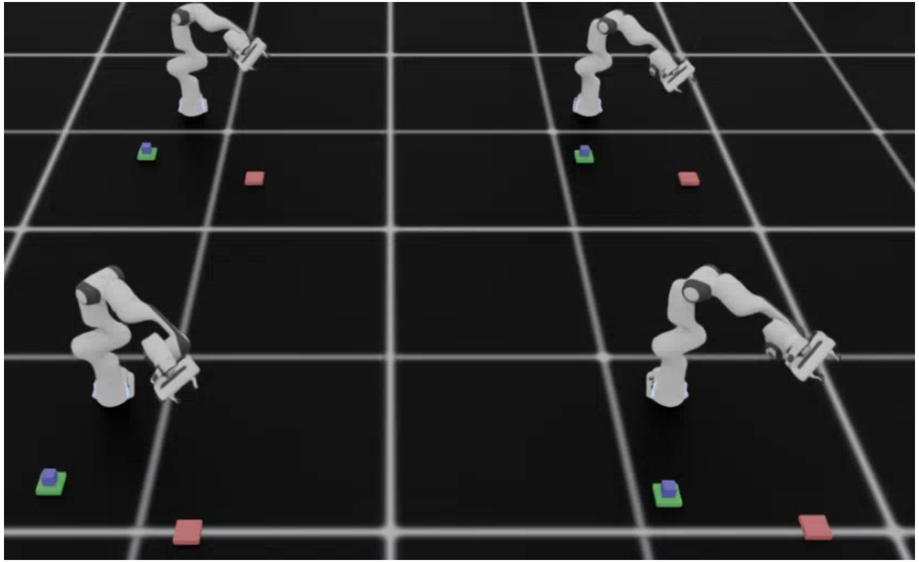
Isaac Sim · Isaac Lab — Extending your midterm with robot learning

**Imitation Learning**

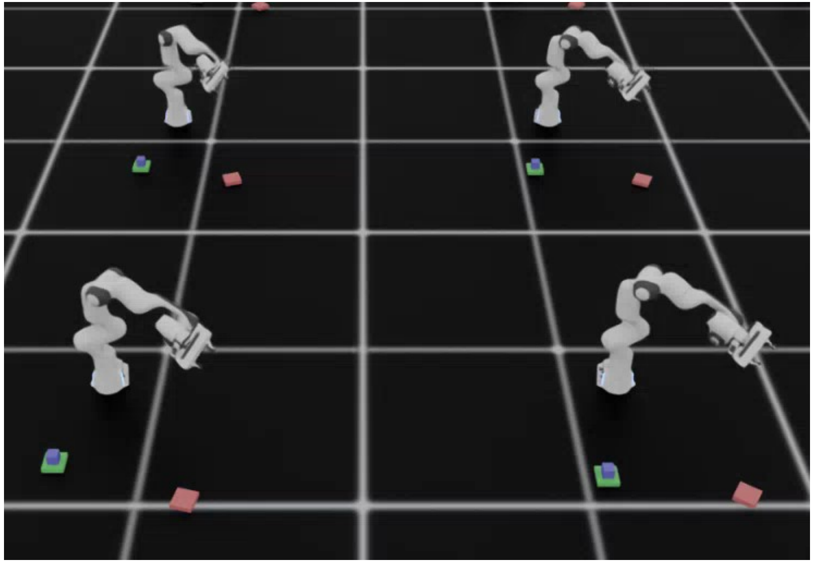
**Reinforcement Learning**

# Final Project Template – IL Task (1/2)

For the final project, we will provide a template that contains all task environments



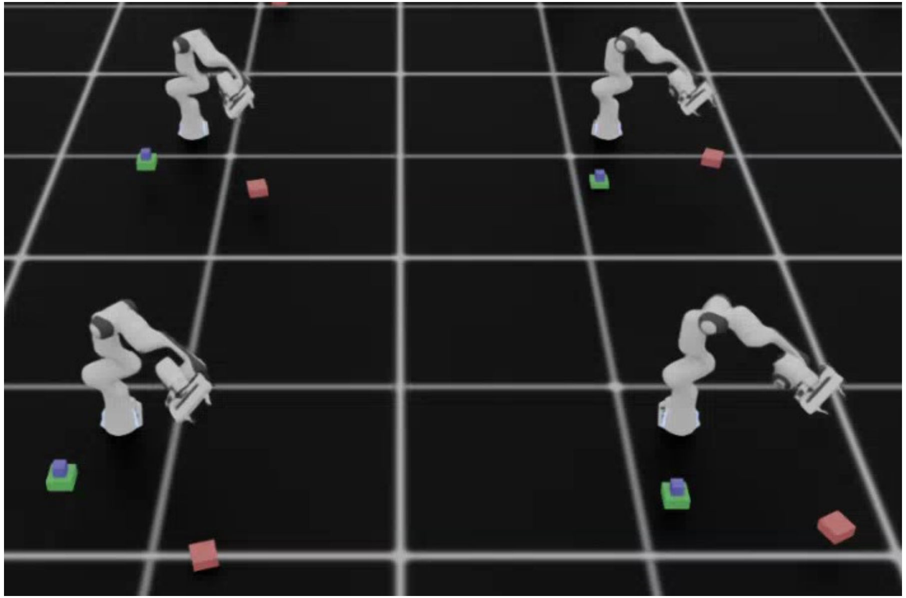
IL Task Level 1
Pick (arbitrary yaw), place (arbitrary yaw)



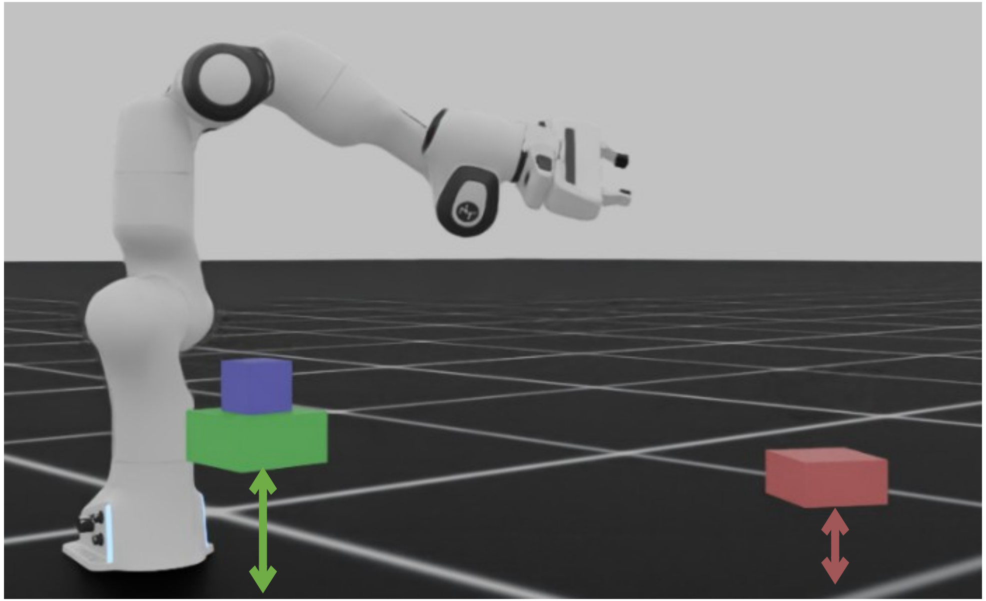
IL Task Level 2
Pick (arbitrary yaw), place (specific yaw)

# Final Project Template – IL Task (2/2)

For the final project, we will provide a template that contains all task environments

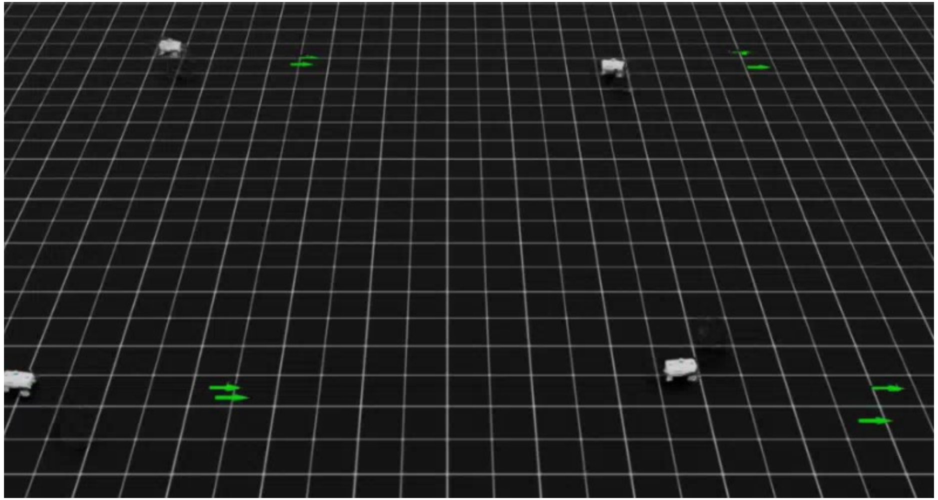


IL Task Level 3
(Based on level2, platform with different height)

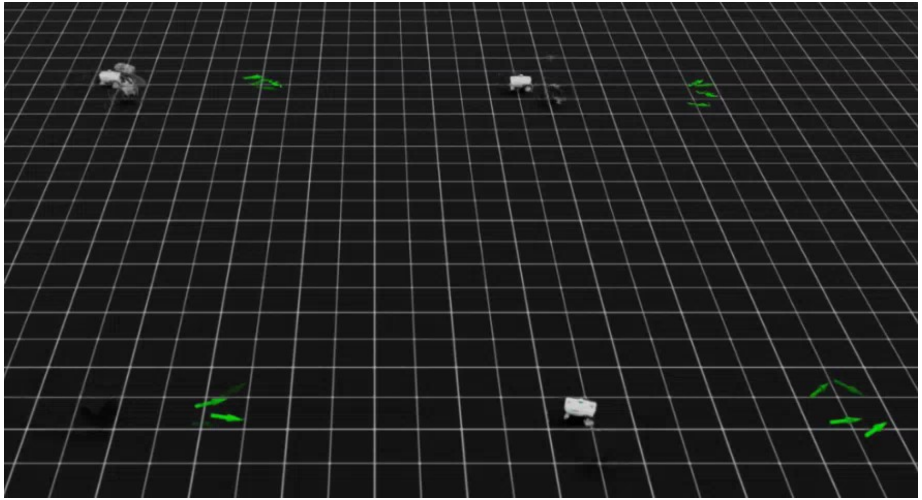


# Final Project Template – RL Task (1/2)

For the final project, we will provide a template that contains all task environments



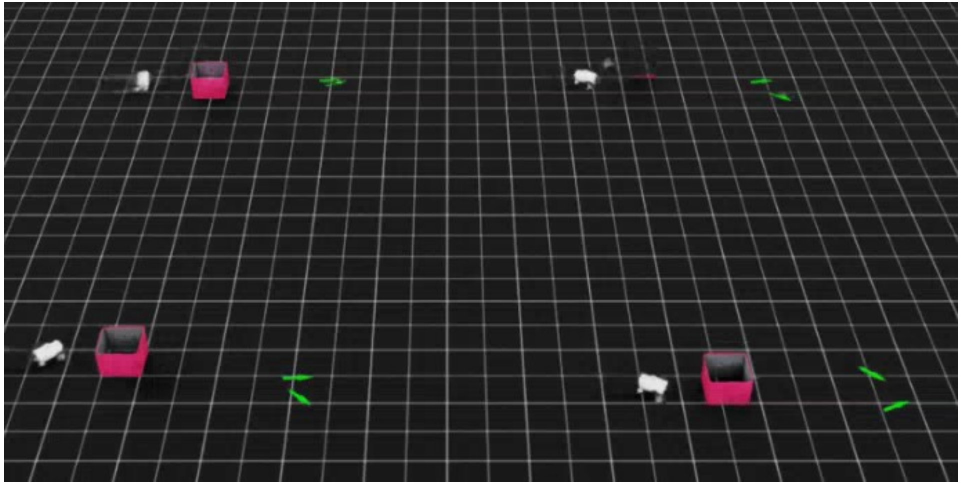
RL Task Level 1
Navigate start to goal (arbitrary yaw)



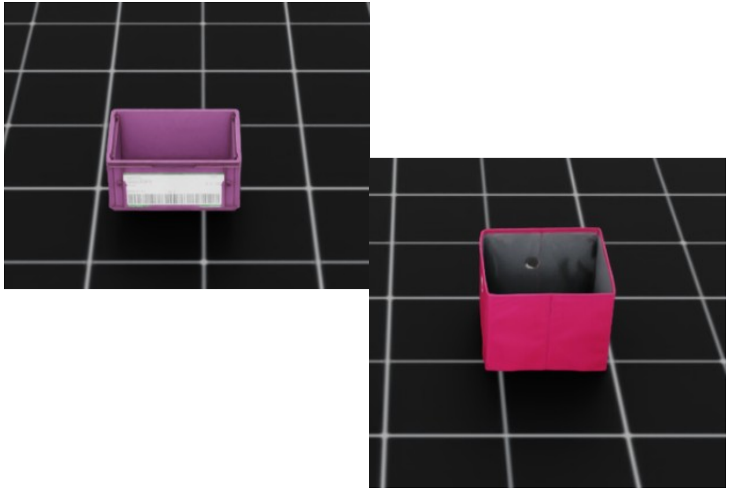
RL Task Level 2
Navigate start to goal (specific yaw)

# Final Project Template – RL Task (2/2)

For the final project, we will provide a template that contains all task environments



RL Task Level 3
Based on Level 2, with obstacle avoidance required



Obstacles will be randomly chosen from the two objects above

# Environmetn List

After registering the environment, run the following command to list all environments in the project

```
python -m pip install -e source/final_project
python scripts/list_envs.py
```

```
+--------------------------------------------------------------------------------------------------------------------------------------------------------------+
|                                                          Available Environments in Isaac Lab                                                                 |
+--------+----------------------+------------------------------------+------------------------------------------------------------------------------------------+
| S. No. | Task Name            | Entry Point                        | Config                                                                                   |
+--------+----------------------+------------------------------------+------------------------------------------------------------------------------------------+
|   1    | FinalProject-IL-L1-v0 | isaaclab.envs:ManagerBasedRLEnv | final_project.tasks.manager_based.final_project.student_interface.IL.env_cfg_L1:EnvCfg |
|   2    | FinalProject-IL-L2-v0 | isaaclab.envs:ManagerBasedRLEnv | final_project.tasks.manager_based.final_project.student_interface.IL.env_cfg_L2:EnvCfg |
|   3    | FinalProject-IL-L3-v0 | isaaclab.envs:ManagerBasedRLEnv | final_project.tasks.manager_based.final_project.student_interface.IL.env_cfg_L3:EnvCfg |
|   4    | FinalProject-RL-L1-v0 | isaaclab.envs:ManagerBasedRLEnv | final_project.tasks.manager_based.final_project.student_interface.RL.env_cfg_L1:EnvCfg |
|   5    | FinalProject-RL-L2-v0 | isaaclab.envs:ManagerBasedRLEnv | final_project.tasks.manager_based.final_project.student_interface.RL.env_cfg_L2:EnvCfg |
|   6    | FinalProject-RL-L3-v0 | isaaclab.envs:ManagerBasedRLEnv | final_project.tasks.manager_based.final_project.student_interface.RL.env_cfg_L3:EnvCfg |
+--------+----------------------+------------------------------------+------------------------------------------------------------------------------------------+
```

# File Structure

```
final_project/
├── 📁 scripts/
└── 📁
source/final_project/tasks/manager_based/final_project/
    ├── 📁 agents/
    │   ├── skrl_ppo_cfg.yaml
    │   └── 📁 robomimic/
    │       └── bc.json
    │
    ├── 📁 env/                       ⚠ DO NOT MODIFY
    │   ├── base_eval_env.py
    │   ├── …
    │   ├── rl_events.py
    │   └── 📁 level_cfgs/
    │       ├── il_L1.py / il_L2.py / il_L3.py
    │       └── rl_L1.py / rl_L2.py / rl_L3.py
    │
    └── 📁 student_interface/         ✅ YOUR WORK
        ├── 📁 IL/
        │   ├── env_cfg_L1.py
        │   ├── env_cfg_L2.py
        │   └── env_cfg_L3.py
        └── 📁 RL/
            ├── env_cfg_L1.py
            ├── env_cfg_L2.py
            └── env_cfg_L3.py
```

- This template is based on the Isaac Lab template generator
- The env folder contains the base environment setup, which is responsible for placing objects and performing randomization
- The student_interface folder contains the environment for training/data collection/validation for **IL** and **RL** tasks, which inherits the base environment

# File Structure (2/2)

```
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source/final_project/tasks/manager_based/final_project/
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    │   └── 📁 robomimic/
    │       └── bc.json
    │
    ├── 📁 env/                        ⚠ DO NOT MODIFY
    │   ├── base_eval_env.py
    │   ├── …
    │   ├── rl_events.py
    │   └── 📁 level_cfgs/
    │       ├── il_L1.py / il_L2.py / il_L3.py
    │       └── rl_L1.py / rl_L2.py / rl_L3.py
    │
    └── 📁 student_interface/          ✅ YOUR WORK
        ├── 📁 IL/
        │   ├── env_cfg_L1.py
        │   ├── env_cfg_L2.py
        │   └── env_cfg_L3.py
        └── 📁 RL/
            ├── env_cfg_L1.py
            ├── env_cfg_L2.py
            └── env_cfg_L3.py
```

- The files are named env_cfg_L{1/2/3}.py, corresponding to different task levels for RL or IL

# Environment Configuration File (1/2)

```python
@configclass
class ObservationsCfg:

    @configclass
    class PolicyCfg(ObsGroup):
        # TODO: add more terms

    policy: PolicyCfg = PolicyCfg()


@configclass
class ActionsCfg:
    # TODO: add more terms
```

env_cfg_L1.py

- The template contains only the minimum runnable environment
- Implement the observation/action configuration (cfg) based on your specific task requirements
- All you need to modify are the environment configuration files corresponding to different tasks

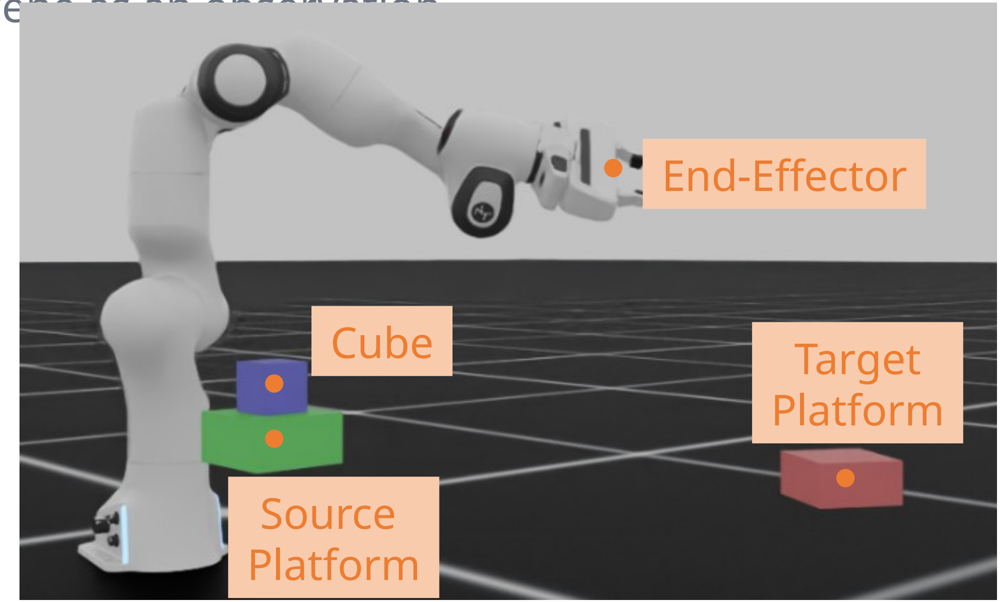
# Environment Configuration File (2/2)

```python
# —— Optional extensions
# Add sensors — subclass ILSceneCfg (has robot, cube, source/target platforms):
...

# Add termination conditions (time_out always present via ILTerminationsCfg):
...


@configclass
class EnvCfg(ILEvalEnvL1Cfg):

    # scene: CustomSceneCfg …
    # terminations: CustomTerminationsCfg …
    …
```

- For students who want to add more sensors, you can inherit the base SceneCfg and add additional setup for the scene
- The same applies to TerminationCfg — inherit the base class and add your custom termination conditions
- There are guides in the comments that you can reference

# Scene Observation Access

- In the observation config, you have full access to the scene to retrieve any information, such as the hand pose, allowing you to directly obtain the current state of the scene as an observation.




Full scene information is available as observation

# Training Environments Evaluation

40 points · You build these yourself — TA provides eval config only

| Environment | Purpose | Points |
|---|---|---|
| **RL training environment** | AMR navigation training | **20** |
| **IL data collection environment** | Expert rollout + dataset generation | **20** |

⚠️ Submitting only the environment class without training evidence = partial credit at best.
Your README must include exact commands to reproduce both environments.

**README must include:**

- Python + Isaac Lab version, any extra dependencies
- Exact command to run IL data collection and training
- Exact command to run RL training

# Trained Policies Evaluation

50 points · Pass threshold: ≥ 80% success rate over evaluation episodes

**IMITATION LEARNING — Franka Pick & Place**

| Level | Task | Pts |
|---|---|---|
| L1 | Pick (arbitrary yaw), place at target<br>Position error < 5 cm | 10 |
| L2 | Extend L1 + specified orientation<br>Yaw error < 15° | 15 |
| L3 ★ Bonus | Pick from randomized-height platform, place at different height with orientation | +5 |

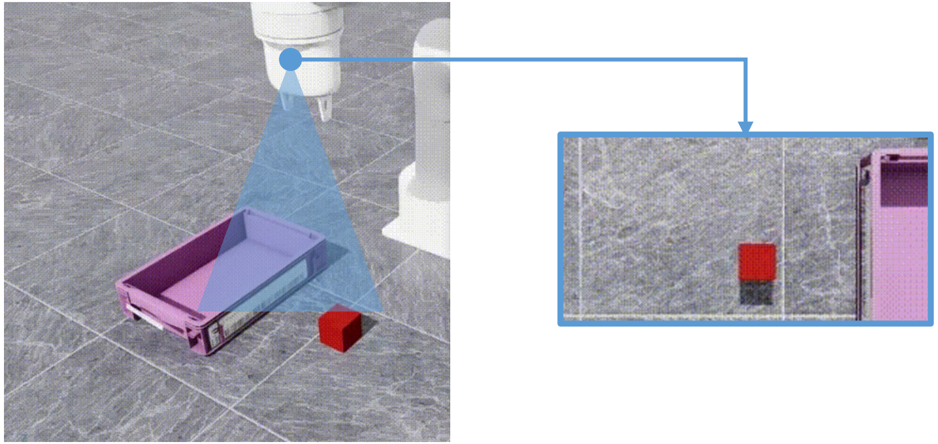
**REINFORCEMENT LEARNING — Nova Carter Navigation**

| Level | Task | Pts |
|---|---|---|
| L1 | Navigate start to goal<br>Position error < 10 cm, straight path | 10 |
| L2 | Extend L1 + specified heading<br>Yaw error < 15° | 15 |
| L3 ★ Bonus | Extend L2 + static obstacles<br>Avoid all collisions<br><br>Obstacle assets provided — evaluation draws from that set only | +10 |

# Bonus Points - Vision-Based Policy

5 points per task

- Bonus points are awarded per task type
  - **5 points per task (L1–L3 for both IL and RL)**
- Use vision as the observation instead of directly accessing scene state
- No direct access to target positions:
  - **IL task**: cube and place target position
  - **RL task**: obstacle and goal position



# Report & Demo Video Evaluation

10 points · PDF max 8 pages + recorded demo in Isaac Sim viewer

### PDF REPORT

- Scene & environment design — assets, sensors, differences from midterm
- IL pipeline — expert policy, demo count, hyperparameters, loss curves
- RL pipeline — obs / action / reward design with rationale, training curves
- Domain randomization — what, ranges, screenshots of variations
- Results — success rate table per level with screenshots / video frames
- Discussion — what worked, what did not, future directions

### DEMO VIDEO

- Record in Isaac Sim viewer
- Show all IL levels attempted
- Show all RL levels attempted
- Label each clip with the level name

Both IL and RL tasks must be demonstrated in the video. Label each segment with its level name for clarity.

# How TAs Will Evaluate the Results

```
final_project/
├── 📁 scripts/
└── 📁
source/final_project/tasks/manager_based/final_project/
    ├── 📁 agents/
    │   ├── skrl_ppo_cfg.yaml
    │   └── 📁 robomimic/
    │       └── bc.json
    │
    ├── 📁 env/                                ⚠ DO NOT MODIFY
    │   ├── base_eval_env.py                     (Replaced by
TAs)
    │   ├── …
    │   ├── rl_events.py
    │   └── 📁 level_cfgs/
    │       ├── il_L1.py / il_L2.py / il_L3.py
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        └── 📁 RL/
            ├── env_cfg_L1.py
            ├── env_cfg_L2.py
```

- When evaluating the results, TAs will replace the base env folder, so do not modify any files in this folder
- TAs will then run the commands provided in the project README file to verify that all functions and tasks work correctly

# Submission Requirements

- Zip your entire project along with the README.
- You may add new files to the template, but the original file name and structure must remain unchanged
- Remove all cache files (e.g., __pycache__) before submitting

# Resources

- In the final project, you are encouraged to try different policies for both IL and RL. The following are some references
  - [Pick and Place Imitation Learning with Isaac Sim & ROS2]
  - [[arXiv 2025] Sim-to-Real Transfer for Mobile Robots with Reinforcement Learni
ng]